AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

SCHOOL OF COMPUTING

Chetan Kumar G

CH.SC.U4CSE24109

OBJECT ORIENTED PROGRAMMING (23CSE111)

LAB RECORD

AMRITA VISHWA VIDYAPEETHAM

AMRITA SCHOOL OF COMPUTING, CHENNAI

BONAFIDE CERTIFICATE

This is to certify that the Lab Record work for 23CSE111- Object Oriented Programming Subject submitted by ***CH.SC.U4CSE24109 – Chetan Kumar G*** in **"Computer Science and Engineering"** is a Bonafide record of the work carried out under my guidance and supervision at Amrita School of Computing, Chennai.

This Lab examination held on

Internal Examiner 1 Internal Examiner 2

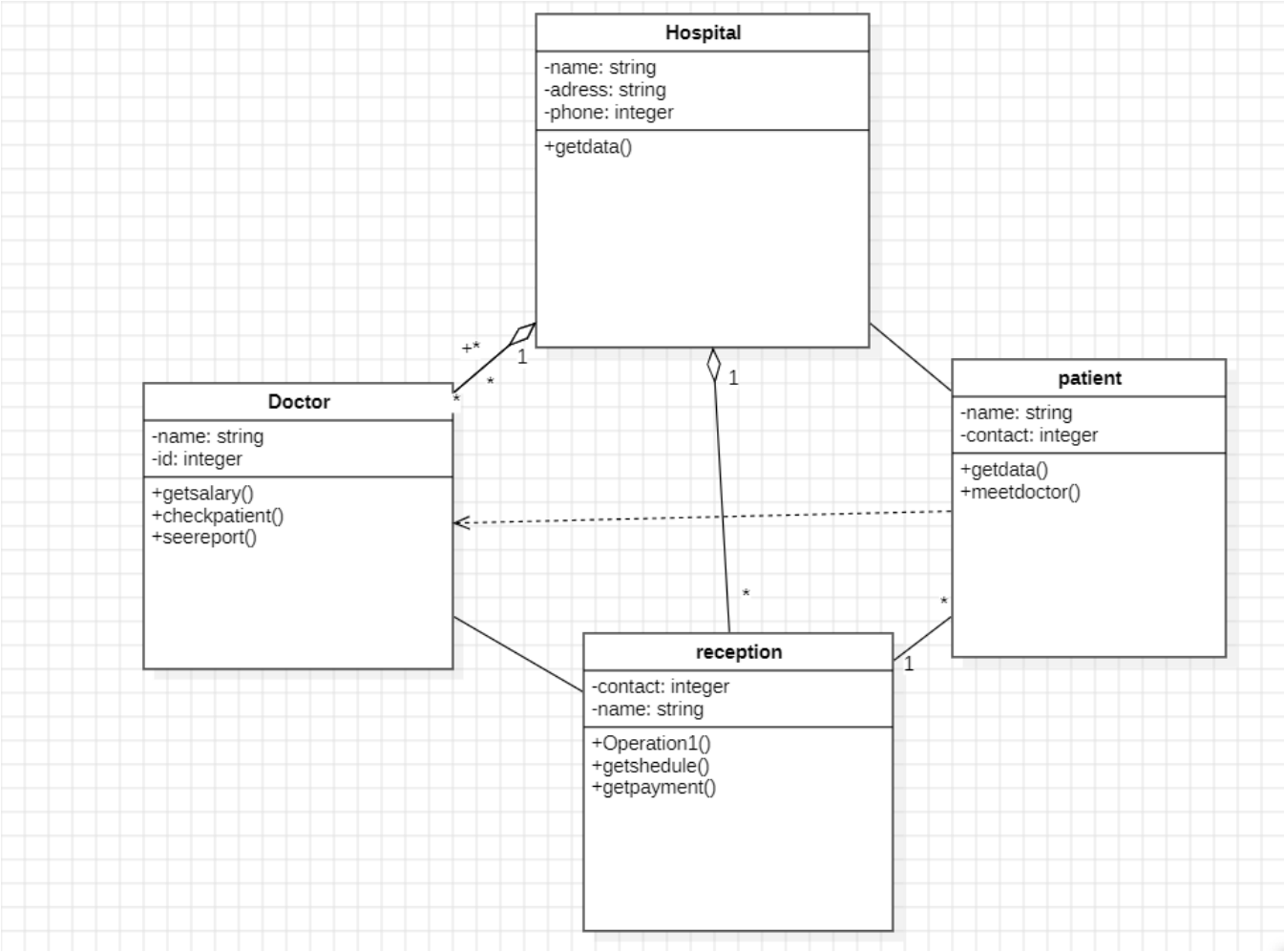
INDEX

S.NO	TITLE	PAGE.NO
	UML DIAGRAM	
1.	**HOSPITAL MANAGEMENT SYSTEM**	
	a) Class Diagram	4
	b) Use Case Diagram	5
	c) Sequence Diagram	6
	d) Object Diagram	7
	e) State-Activity Diagram	8
2.	**ONLINE FOOD ORDERING SYSTEM**	
	a) Class Diagram	9
	b) Use Case Diagram	10
	c) Sequence Diagram	11
	d) Object Diagram	12
	e) State-Activity Diagram	13
3.	**BASIC JAVA PROGRAMS**	
	a) Check Even or Odd	14
	b) Largest of Three Numbers	15
	c) Reverse a Number	17
	d) Sum of Digits	18
	e) Print Fibonacci Series	19
	f) Prime Number Check	20
	g) Print a Pyramid Pattern	21
	h) Factorial of a Number	22
	i) Armstrong Number	23
	j) Number Guessing Game	24

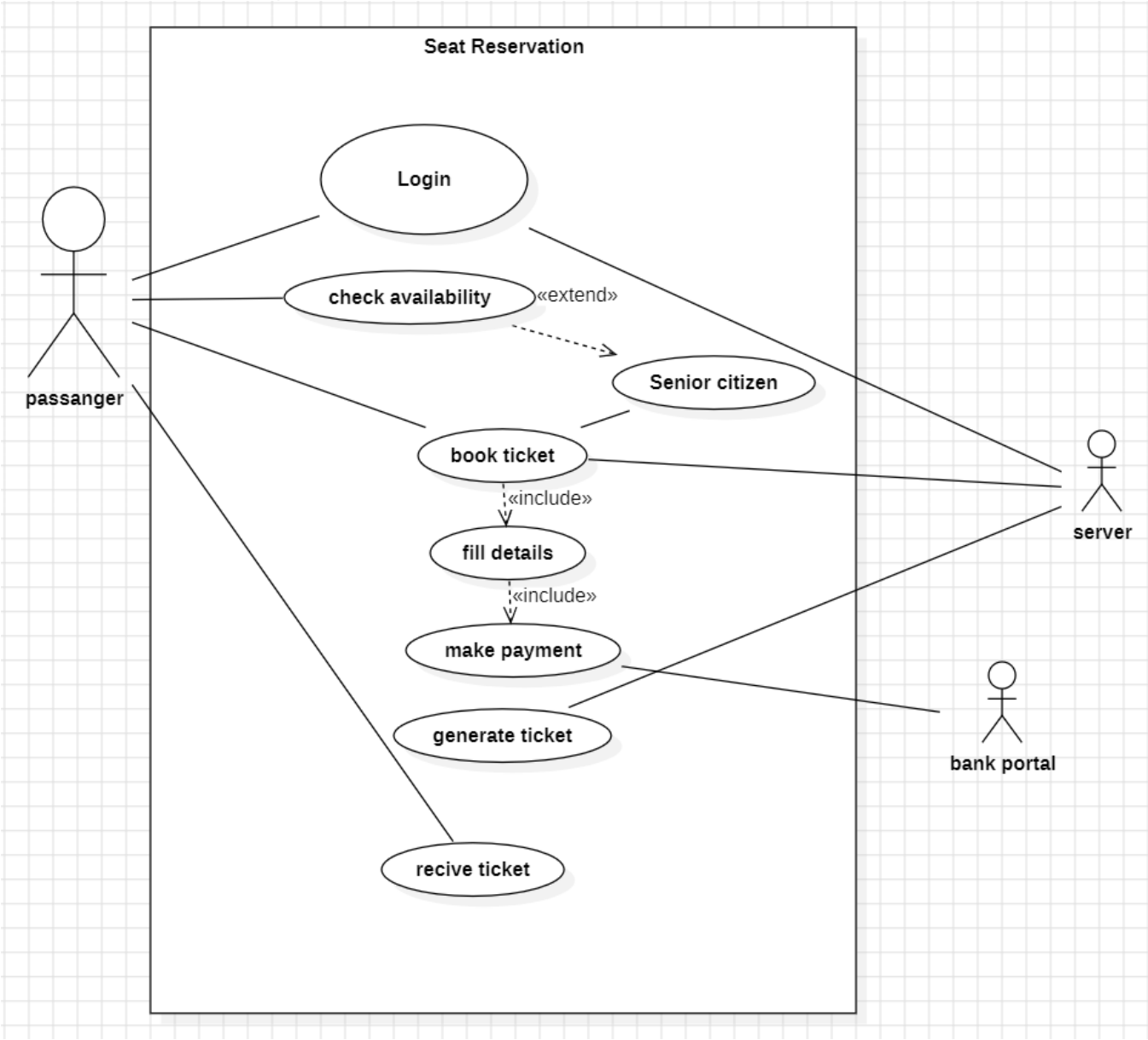
UML DIAGRAMS

1. HOSPITAL MANAGEMENT SYSTEM

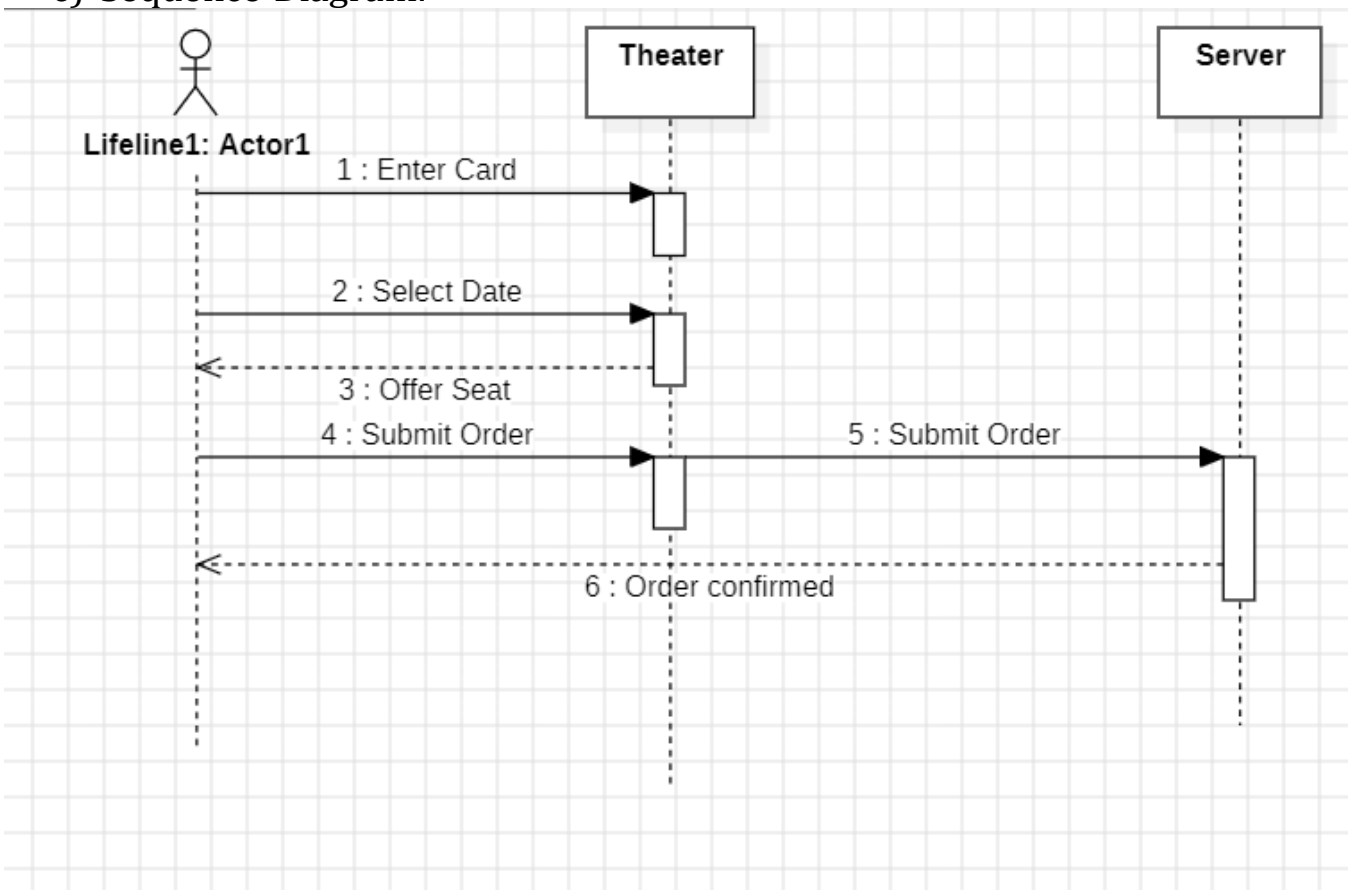
a) Class Diagram



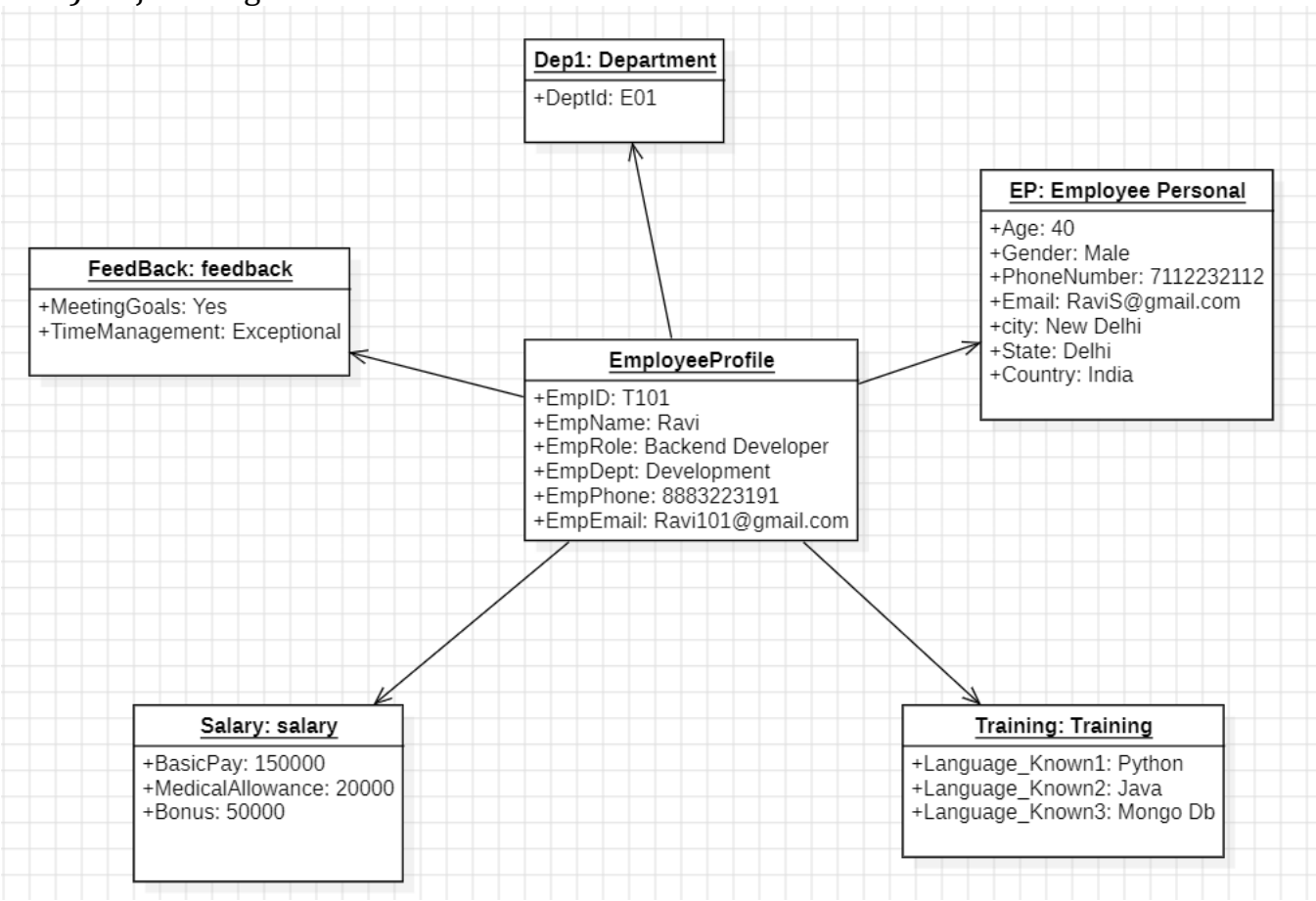
b) Use Case Diagram:



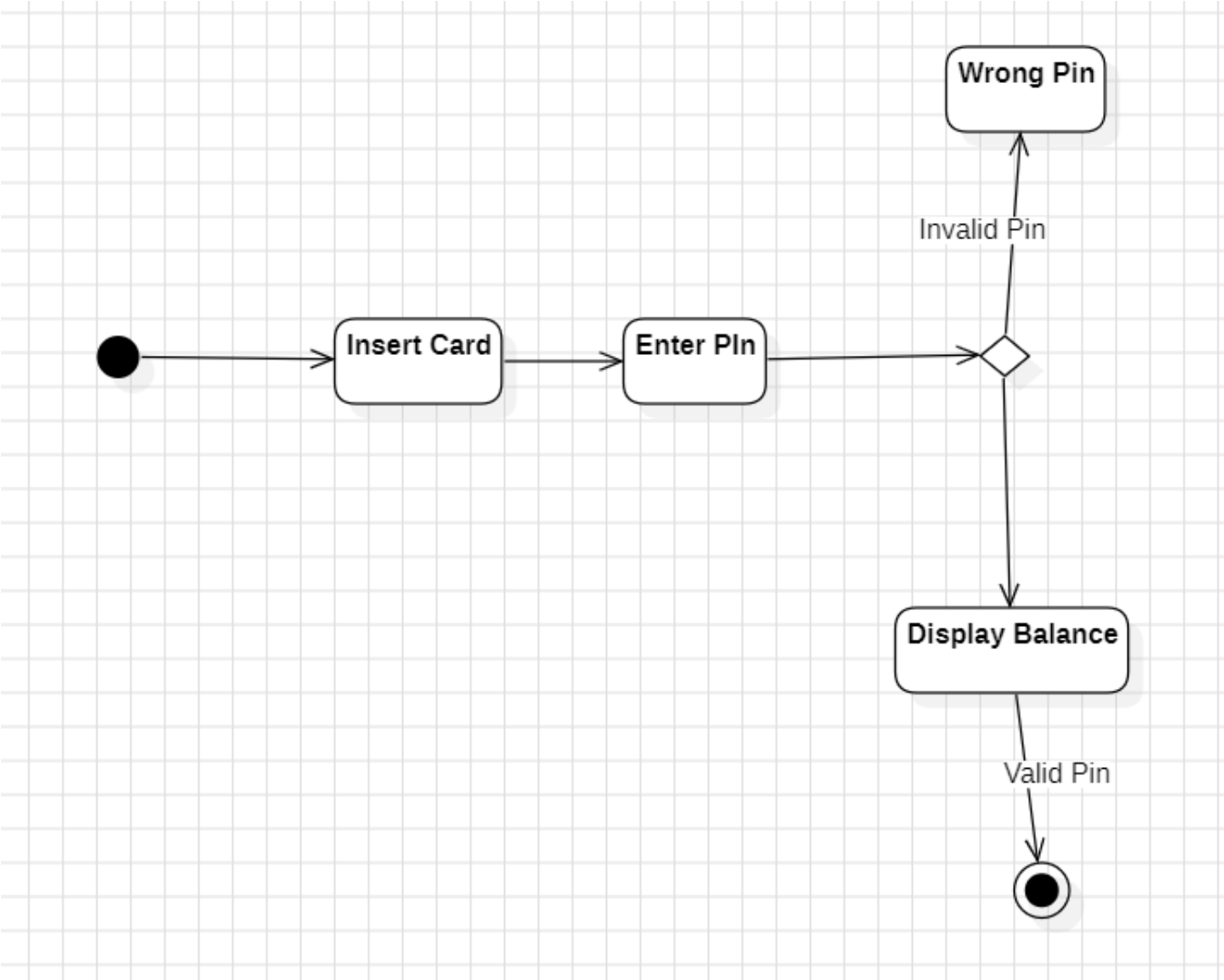
c) Sequence Diagram:



d) Object Diagram:

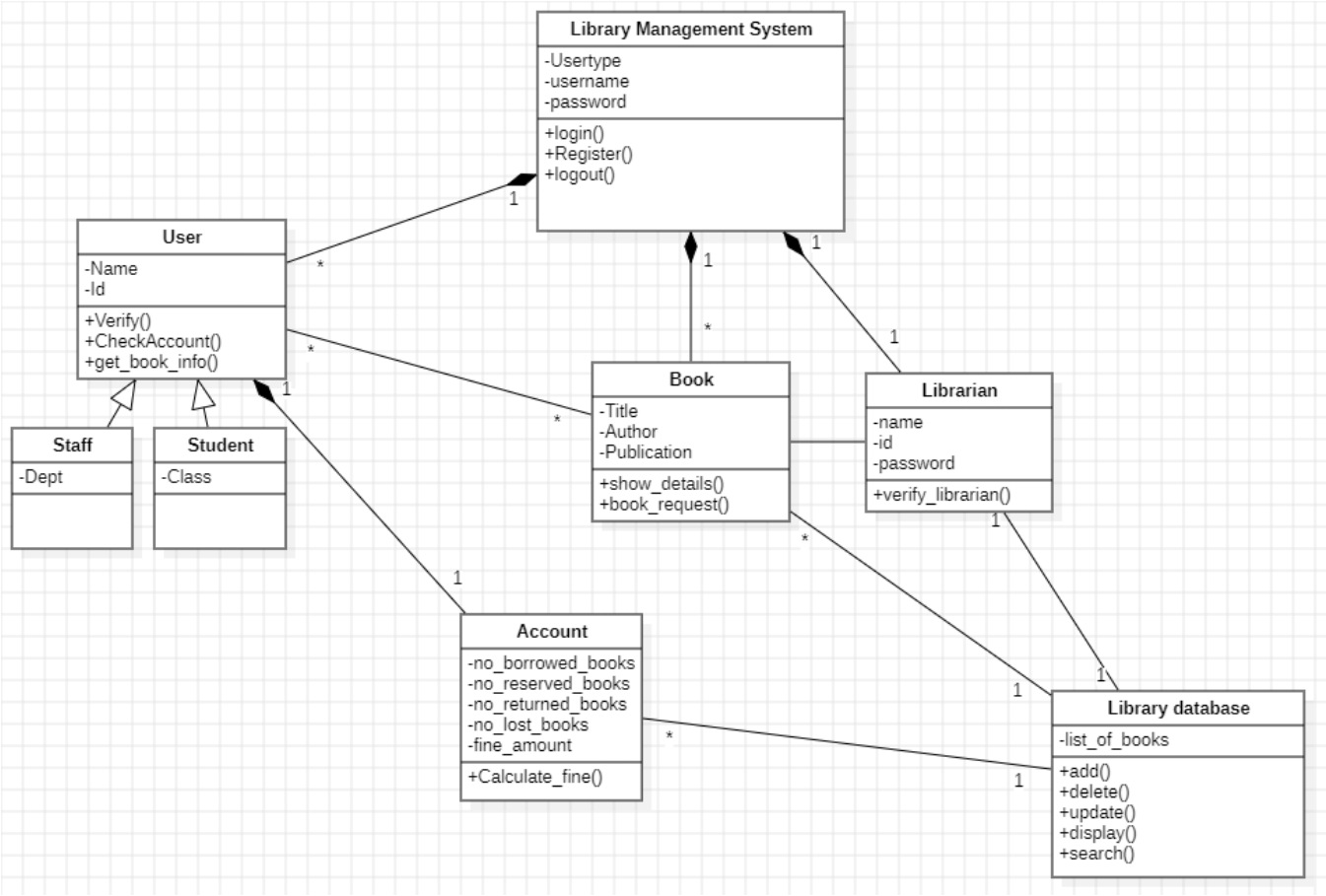


e) State-Activity Diagram:

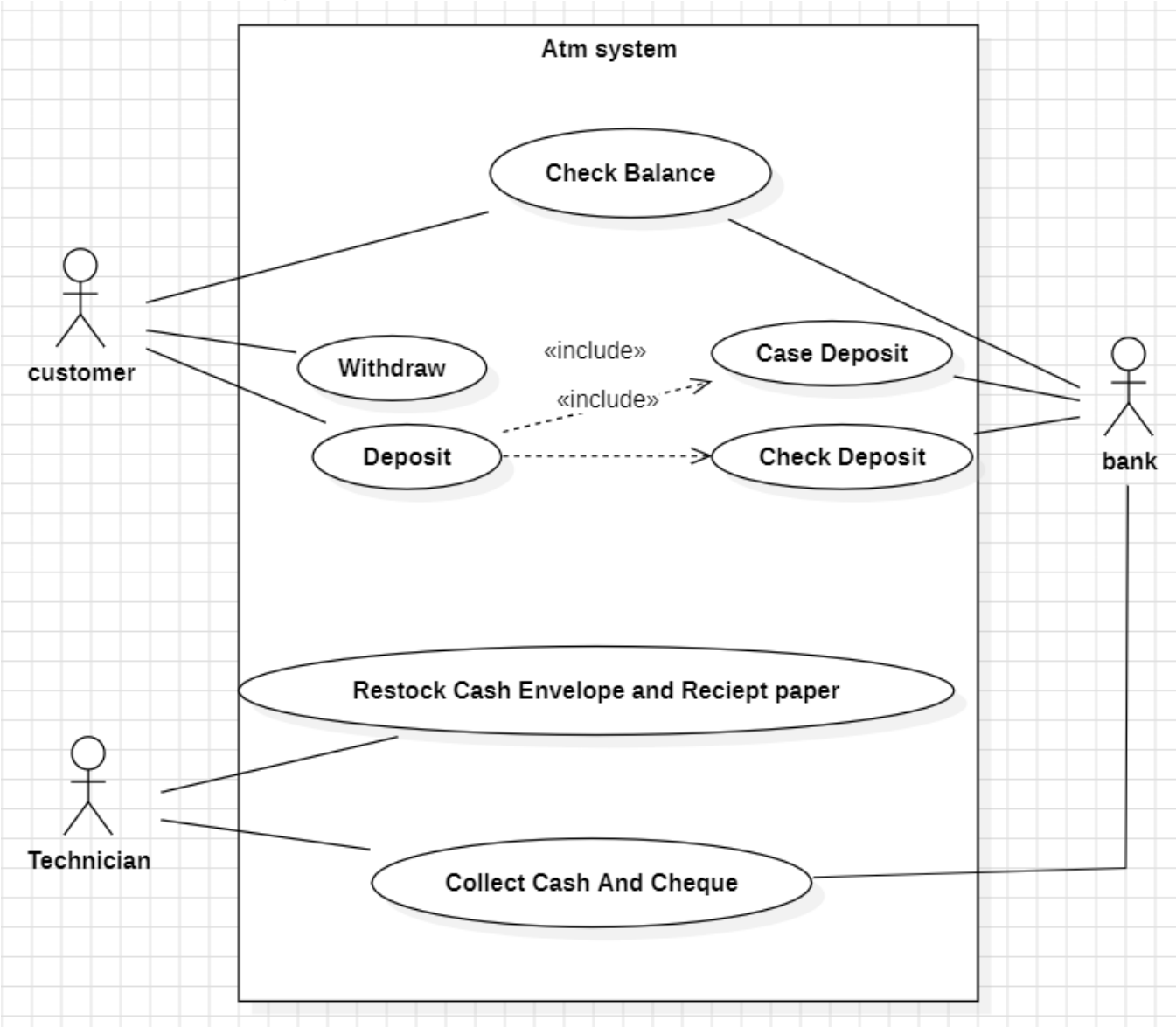


2.ONLINE FOOD ORDERING SYSTEM

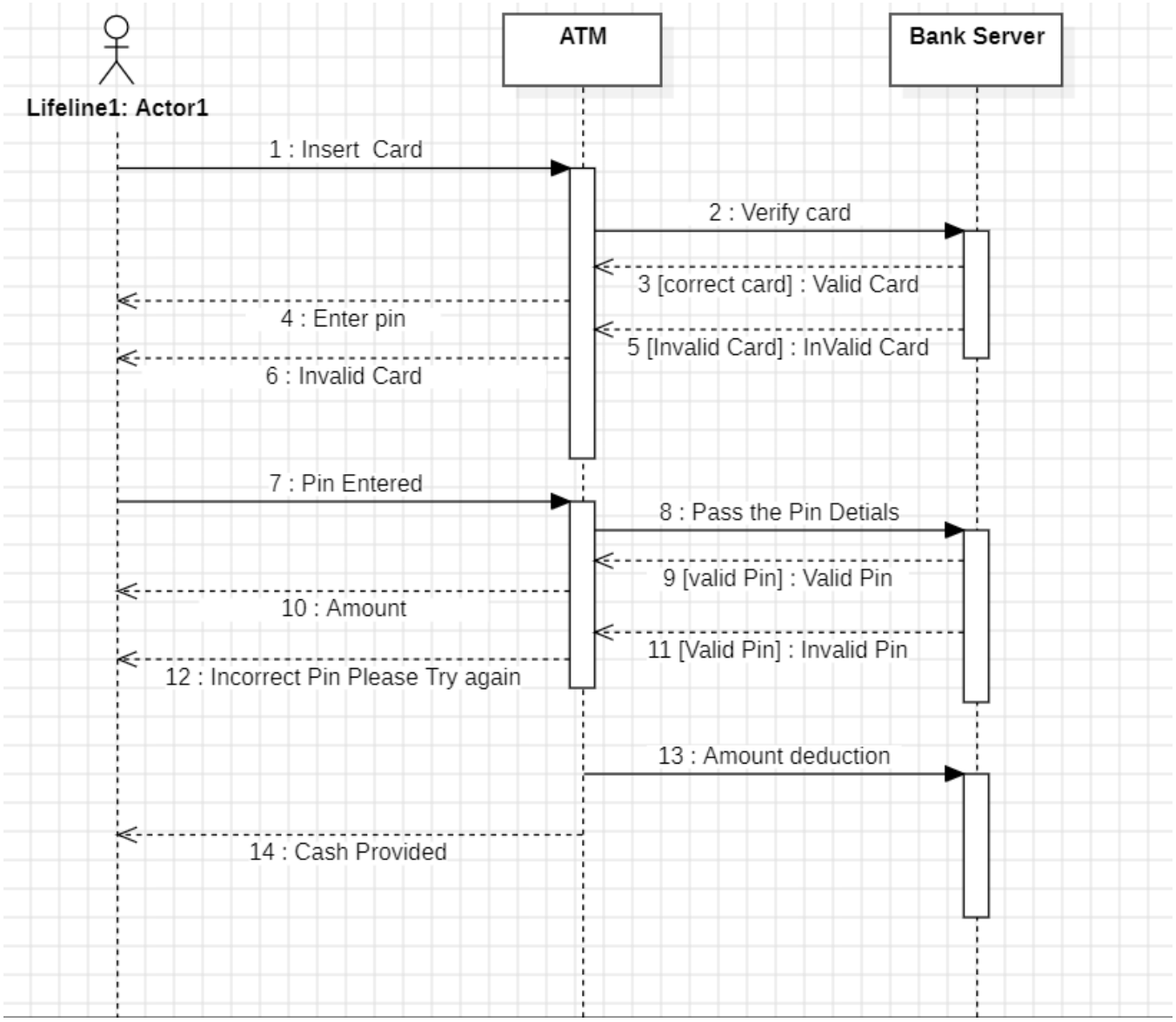
a) Class Diagram:



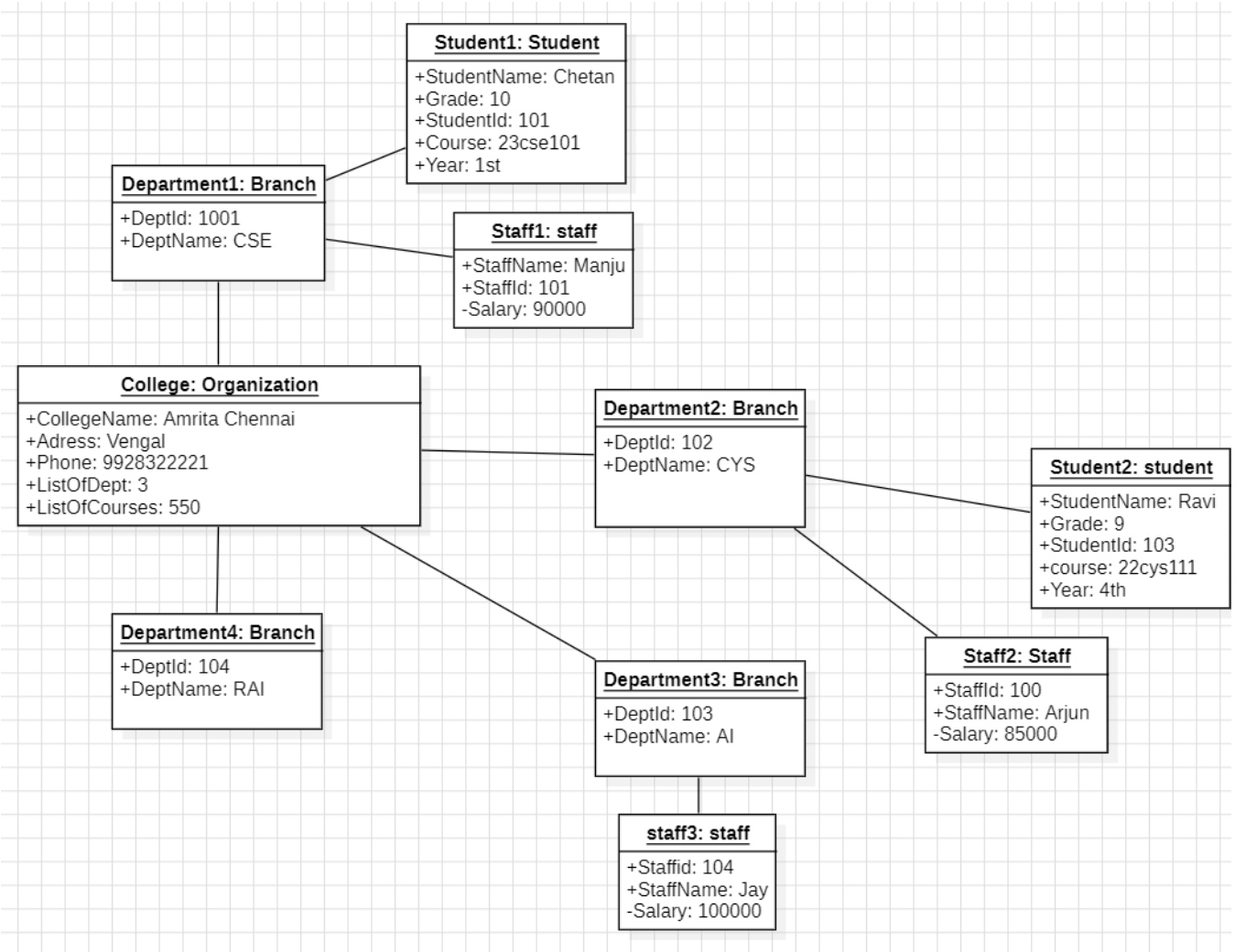
b) Use Case Diagram:



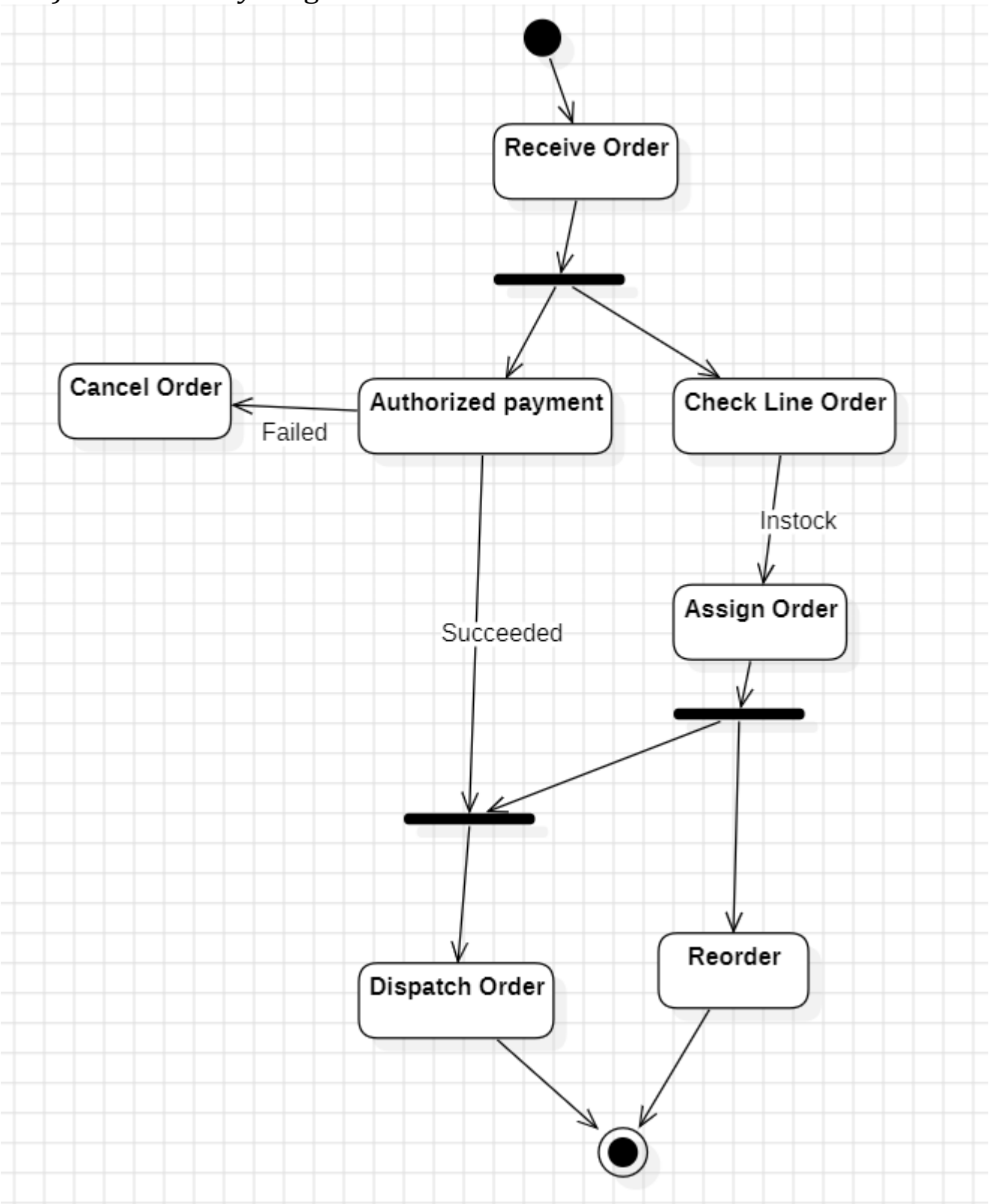
c) Sequence Diagram:



d) Object Diagram:



e) State-Activity Diagram:



3.Basic Java Programs

a) Check Even or Odd

Write a Java program that takes an integer input from the user and determines whether it is even or odd using an if-else statement.

Code:

```
import java.util.*;
public class oddeven{

      public static void main(String[] args){
            Scanner obj = new Scanner(System.in);
            System.out.print("Enter a Number-");
            int num = obj.nextInt();

            if(num%2==0){
                  System.out.println("the number is even");
            }

            else{
                  System.out.println("the number is odd");
            }
      }
}
```

Output:

```
D:\java>javac oddeven.java

D:\java>java oddeven
Enter a Number-3
the number is odd
```

b) Largest of Three Numbers

Write a program that takes three integers as input and determines the largest among them using nested if-else statements.

Code:

```
import java.util.*;
public class largest{

      public static void main(String[] args){
            Scanner obj = new Scanner(System.in);
            System.out.print("Enter first Number-");
            int num1 = obj.nextInt();
            System.out.print("Enter second Number-");
            int num2 = obj.nextInt();
            System.out.print("Enter third Number-");
            int num3 = obj.nextInt();

            if(num1>num2){
                  if(num1>num3){
                        System.out.println("the First number is Greatest" + num1);
                  }
                  else if(num3>num1){
                        System.out.println("the Third number is Greatest" + num3);
                  }
                  else{
                        System.out.println("Two numbers are same");
                  }
            }

            else if(num2>num1){
                  if(num2>num3){
                        System.out.println("the Second number is Greatest" + num2);
                  }
                  else if(num3>num2){
                        System.out.println("the Third number is Greatest" + num3);
                  }
                  else{
                        System.out.println("Two numbers are same");

                  }

            }
      }
}
```

Output:

```
D:\java>javac largest.java

D:\java>java largest
Enter first Number-2
Enter second Number-3
Enter third Number-1
the Second number is Greatest3
```

c) Reverse a Number

Write a Java program that takes an integer input and reverses its digits using a while loop.
Example:
Input: 1234
Output: 4321

Code:
```
import java.util.*;
public class reverse{

      public static void main(String[] args){
            Scanner obj = new Scanner(System.in);
            System.out.print("Enter a Number-");
            int num = obj.nextInt();
            int reverse = 0;

            while (num!=0){
                  int digit=num%10;
                  reverse=reverse*10+digit;
                  num=num/10;
            }
            System.out.println("the reverse digit is " + reverse);
      }
}
```

Output:

```
D:\java>javac reverse.java

D:\java>java reverse
Enter a Number-123
the reverse digit is 321
```

d) Sum of Digits

Write a program to find the sum of the digits of a given number using a while loop.
Example:
Input: 145
Output: 10 (1+4+5)

Code:
```
import java.util.*;
public class sumOfDigits{

      public static void main(String[] args){
            Scanner obj = new Scanner(System.in);
            System.out.print("Enter a Number-");
            int num = obj.nextInt();
            int sum= 0;

            while (num!=0){
                  int digit=num%10;
                  sum=sum+digit;
                  num=num/10;
            }
            System.out.println("the reverse digit is " + sum);
      }
}
```

Output;

```
D:\java>javac sumOfDigits.java

D:\java>java sumOfDigits
Enter a Number-123
the reverse digit is 6
```

e) Print Fibonacci Series

Write a Java program to print the first n numbers of the Fibonacci series using a for loop.
Example:
Input: 5
Output: 0 1 1 2 3

Code:
```
import java.util.*;
public class fibonacciSeries{

      public static void main(String[] args){
            Scanner obj = new Scanner(System.in);
            System.out.print("Enter a Number-");
            int num = obj.nextInt();
            int a= 0;
            int b= 1;
            System.out.print(a + " ");
            System.out.print(b + " ");
            num=num-2;
            while (num!=0){
                  int sum= a+b;
                  System.out.print(sum + " ");
                  a=b;
                  b=sum;
                  num=num-1;
            }
      }
}
```

Output:

```
D:\java>javac fibonacciSeries.java

D:\java>java fibonacciSeries
Enter a Number-5
0 1 1 2 3
```

f) Prime Number Check

Write a Java program that takes an integer input and checks whether it is a prime number or not using a for loop.

Code:

```
import java.util.*;
public class primeNumber{

      public static void main(String[] args){
            Scanner obj = new Scanner(System.in);
            System.out.print("Enter a Number-");
            int num = obj.nextInt();
            for (int i=2; i<num; i++){
                  if(num%i==0){
                        System.out.println("the number is not prime");
                        break;
                  }
                  else{
                        System.out.println("the number is prime");
                  }
            }
      }
}
```

Output:

```
D:\java>javac primeNumber.java

D:\java>java primeNumber
Enter a Number-3
the number is prime
```

g) Print a Pyramid Pattern

Write a program that prints a pyramid pattern of * based on user input.
Example:
Input: 5
Output:

```
        *
       ***
      *****
     *******
    *********
```

Code:

```
import java.util.*;
public class pattern{

      public static void main(String[] args){
            Scanner obj = new Scanner(System.in);
            System.out.print("Enter a Number-");
            int num = obj.nextInt();
            for (int i=1; i<=num; i++){
                  for(int k=0; k<=num-i; k++){
                        System.out.print(" ");
                  }
                  for(int j=1; j<=(2*i-1); j++){
                        System.out.print("*");
                  }
                  System.out.println();
            }
      }
}
```

Output:

```
D:\java>javac pattern.java

D:\java>java pattern
Enter a Number-5
      *
     ***
    *****
   *******
  *********
```

h) Factorial of a Number

Write a Java program that calculates the factorial of a given number using a for loop.

Example:
Input: 5
Output: 120 (5! = 5 × 4 × 3 × 2 × 1)

Code:

```
import java.util.*;
public class factorial{

      public static void main(String[] args){
            Scanner obj = new Scanner(System.in);
            System.out.print("Enter a Number-");
            int num = obj.nextInt();
            int factorial=1;
            for (int i=1; i<=num; i++){
                  factorial=factorial*i;
            }
            System.out.print(factorial);
      }
}
```

Output:

```
D:\java>javac factorial.java

D:\java>java factorial
Enter a Number-5
120
```

i) Armstrong Number

Write a Java program to check if a number is an Armstrong number (sum of cubes of digits equals the number).
Example:
Input: 153
Output: Yes, it's an Armstrong number.
(Because $1^3 + 5^3 + 3^3 = 153$)

Code:

```
import java.util.*;
public class armstrongNumber{

      public static void main(String[] args){
            Scanner obj = new Scanner(System.in);
            System.out.print("Enter a Number-");
            int num = obj.nextInt();
            int num2=num;
            int sum= 0;
            String str = String.valueOf(num);
            int len = str.length();

            while (num!=0){
                  int digit=num%10;
                  sum=sum+(int) Math.pow(digit, len);
                  num=num/10;
            }
            if(sum==num2){
                  System.out.println("this is an Armstrong number");
            }
            else{
                        System.out.println("this is not an Armstrong number");
            }
      }
}
```

Output:

```
D:\java>javac armstrongNumber.java

D:\java>java armstrongNumber
Enter a Number-153
this is an Armstrong number
```

j) Number Guessing Game

Write a simple number guessing game where the program randomly selects a number (1-100), and the user keeps guessing until they get it right.

Code:

```
import java.util.*;
import java.util.Random;
public class guessNumber{

      public static void main(String[] args){
            Random random= new Random();
            int rannum = random.nextInt(100)+1;
            Scanner obj = new Scanner(System.in);
            Boolean a=true;
            while(a){
                  System.out.print("Enter a Number-");
                  int num = obj.nextInt();
                  if (num==rannum){
                        System.out.print("Congratulation!!! Correct");
                        a=false;
                  }
                  else{
                        System.out.println("Oops..., try again");
                  }
            }
      }
}
```

Output:

```
D:\java>javac guessNumber.java

D:\java>java guessNumber
Enter a Number-32
Oops..., try again
Enter a Number-43
Oops..., try again
Enter a Number-52
Congratulation!!! Correct
```